4.5V to 18V Input, 4-A Synchronous Step-Down SWIFT™ Converter with Eco-Mode™

Check for Samples: TPS54426

FEATURES

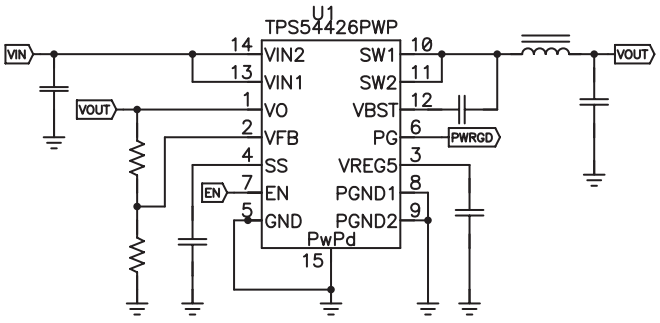
- **D-CAP2™ Mode Enables Fast Transient Response**
- **Low Output Ripple and Allows Ceramic Output Capacitor**
- **Wide V_{IN} Input Voltage Range: 4.5 V to 18 V**
- **Output Voltage Range: 0.76 V to 5.5 V**
- **Highly Efficient Integrated FET's Optimized for Lower Duty Cycle Applications**
  – 65 mΩ (High Side) and 55 mΩ (Low Side)
- **High Efficiency, less than 10 μA at shutdown**
- **High Initial Bandgap Reference Accuracy**
- **Adjustable Soft Start**
- **Pre-Biased Soft Start**
- **700-kHz Switching Frequency (f_{SW})**
- **Cycle By Cycle Over Current Limit**
- **Power Good Output**
- **Auto-Skip Eco-mode™ for High Efficiency at Light Load**

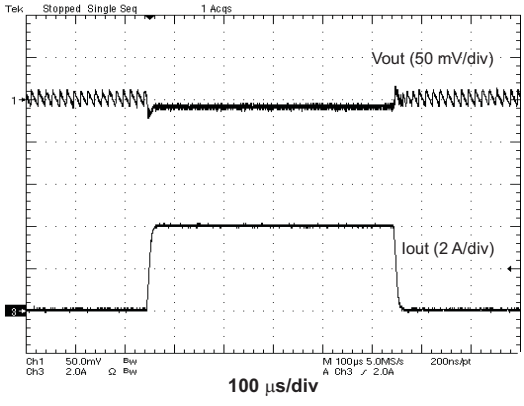
APPLICATIONS

- **Wide Range of Applications for Low Voltage System**
  - **Digital TV Power Supply**
  - **High Definition Blu-ray Disc™ Players**
  - **Networking Home Terminal**
  - **Digital Set Top Box (STB)**

DESCRIPTION

The TPS54426 is an adaptive on-time D-CAP2™ mode synchronous buck converter. The TPS54426 enables system designers to complete the suite of various end equipment's power bus regulators with a cost effective, low component count, low standby current solution. The main control loop for the TPS54426 uses the D-CAP2™ mode control which provides a very fast transient response with no external compensation components. The adaptive on-time control supports seamless transition between PWM mode at higher load conditions and Eco-mode™ operation at light loads. Eco-mode™ allows the TPS54426 to maintain high efficiency during lighter load conditions. The TPS54426 also has a proprietary circuit that enables the device to adopt to both low equivalent series resistance (ESR) output capacitors, such as POSCAP, SP-CAP, and ultra-low ESR ceramic capacitors. The device operates from 4.5-V to 18-V VIN input. The output voltage can be programmed between 0.76 V and 5.5 V. The device also features an adjustable soft start time and a power good function. The TPS54426 is available in the 14-pin HTSSOP package and the 16 pin QFN package, designed to operate from –40°C to 85°C.





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This integrated circuit can be damaged by ESD. Texas Instruments recommends that all integrated circuits be handled with appropriate precautions. Failure to observe proper handling and installation procedures can cause damage.

ESD damage can range from subtle performance degradation to complete device failure. Precision integrated circuits may be more susceptible to damage because very small parametric changes could cause the device not to meet its published specifications.

ORDERING INFORMATION(1)

T_A	PACKAGE(2) (3)	ORDERABLE PART NUMBER	PIN	TRANSPORT MEDIA
–45°C to 85°C	PowerPAD™ (HTSSOP) – PWP	TPS54426PWP	14	Tube
		TPS54426PWPR		Tape and Reel
	Plastic Quad Flat Pack (QFN)	TPS54426RSAT	16	Tape and Reel
		TPS54426RSAR		Tape and Reel

(1) For the most current package and ordering information, see the Package Option Addendum at the end of this document, or see the TI web site at www.ti.com.
(2) Package drawings, thermal data, and symbolization are available at www.ti.com/packaging.
(3) All package options have Cu NIPDAU lead/ball finish.

ABSOLUTE MAXIMUM RATINGS

over operating free-air temperature range (unless otherwise noted) (1)

			VALUE	UNIT
V_I	Input voltage range	V_{IN1}, V_{IN2}, EN	–0.3 to 20	V
		V_{BST}	–0.3 to 26	V
		V_{BST} (10 ns transient)	–0.3 to 28	V
		V_{FB}, V_O, SS, PG	–0.3 to 6.5	V
		SW1, SW2	–2 to 20	V
		SW1, SW2 (10 ns transient)	–3 to 22	V
V_O	Output voltage range	V_{REG5}	–0.3 to 6.5	V
		P_{GND1}, P_{GND2}	–0.3 to 0.3	V
V_{diff}	Voltage from GND to POWERPAD		–0.2 to 0.2	V
ESD rating	Electrostatic discharge	Human Body Model (HBM)	2	kV
		Charged Device Model (CDM)	500	V
T_J	Operating junction temperature		–40 to 150	°C
T_{stg}	Storage temperature		–55 to 150	°C

(1) Stresses beyond those listed under *absolute maximum ratings* may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated under *recommended operating conditions* is not implied. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.

THERMAL INFORMATION

THERMAL METRIC(1)		TPS54426		UNITS
		PWP (14) PINS	RSA (16) PINS	
θ_{JA}	Junction-to-ambient thermal resistance	55.6	37.3	°C/W
θ_{JCtop}	Junction-to-case (top) thermal resistance	51.3	42.5	
θ_{JB}	Junction-to-board thermal resistance	36.4	14.9	
ψ_{JT}	Junction-to-top characterization parameter	1.8	0.8	
ψ_{JB}	Junction-to-board characterization parameter	20.6	14.8	
θ_{JCbot}	Junction-to-case (bottom) thermal resistance	4.3	4.9	

(1) For more information about traditional and new thermal metrics, see the *IC Package Thermal Metrics* application report, SPRA953.

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RECOMMENDED OPERATING CONDITIONS

over operating free-air temperature range (unless otherwise noted)

			MIN	MAX	UNIT
V_{IN}	Supply input voltage range		4.5	18	V
V_I	Input voltage range	VBST	–0.3	24	V
		VBST (10 ns transient)	–0.3	27	
		SS, PG	–0.1	5.7	
		EN	–0.1	18	
		VO, VFB	–0.1	5.5	
		SW1, SW2	–1.8	18	
		SW1, SW2 (10 ns transient)	–3	21	
		PGND1, PGND2	–0.1	0.1	
V_O	Output voltage range	VREG5	–0.1	5.7	V
I_O	Output Current range	I_{VREG5}	0	10	mA
T_A	Operating free-air temperature		–40	85	°C
T_J	Operating junction temperature		–40	150	°C

ELECTRICAL CHARACTERISTICS

over operating free-air temperature range, V_{IN} = 12V (unless otherwise noted)

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
SUPPLY CURRENT						
I_{VIN}	Operating - non-switching supply current	V_{IN} current, T_A = 25°C, EN = 5 V, V_{FB} = 0.8 V		850	1300	µA
I_{VINSDN}	Shutdown supply current	V_{IN} current, T_A = 25°C, EN = 0 V		1.8	10	µA
LOGIC THRESHOLD						
V_{ENH}	EN high-level input voltage	EN	2			V
V_{ENL}	EN low-level input voltage	EN			0.4	V
V_{FB} VOLTAGE AND DISCHARGE RESISTANCE						
V_{FBTH}	V_{FB} threshold voltage	VFB voltage light load mode, T_A = 25°C, V_O = 1.05 V, I_O = 10mA		771		mV
		T_A = 25°C, V_O = 1.05 V, continuous mode	757	765	773	
		T_A = 0°C to 85°C, V_O = 1.05 V, continuous mode(1)	753		777	
		T_A = –40°C to 85°C, V_O = 1.05 V, continuous mode(1)	751		779	
I_{VFB}	V_{FB} input current	V_{FB} = 0.8 V, T_A = 25°C		0	±0.1	µA
R_{Dischg}	V_O discharge resistance	EN = 0 V, V_O = 0.5 V, T_A = 25°C		50	100	Ω
V_{REG5} OUTPUT						
V_{VREG5}	V_{REG5} output voltage	T_A = 25°C, 6 V < V_{IN} < 18 V, 0 < I_{VREG5} < 5 mA	5.3	5.5	5.7	V
V_{LN5}	Line regulation	6.0 V < V_{IN} < 18 V, I_{VREG5} = 5 mA			20	mV
V_{LD5}	Load regulation	0 mA < I_{VREG5} < 5 mA			100	mV
I_{VREG5}	Output current	V_{IN} = 6 V, V_{REG5} = 4 V, T_A = 25°C		70		mA
MOSFET						
R_{dsonh}	High side switch resistance	25°C, V_{BST} - SW1,2 = 5.5 V		63		mΩ
R_{dsonl}	Low side switch resistance	25°C		55		mΩ

(1) Not production tested.

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ELECTRICAL CHARACTERISTICS (continued)

over operating free-air temperature range, V_{IN} = 12V (unless otherwise noted)

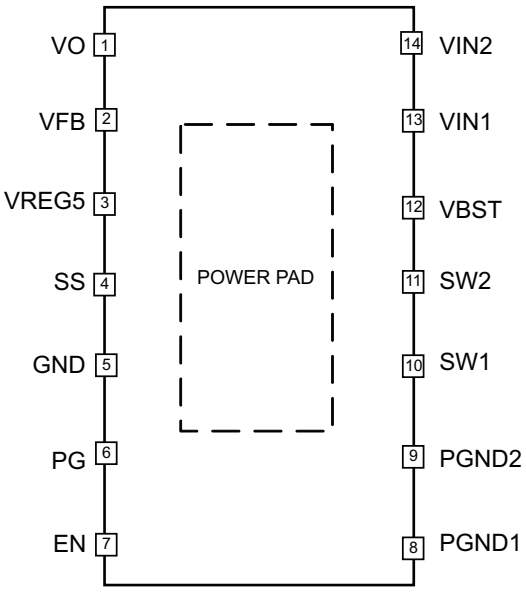
PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
CURRENT LIMIT						
I_{ocl}	Current limit	L_{OUT} = 1.5 µH(2), T_A = -20 ºC to 85 ºC	4.7	5.4	7.5	A
THERMAL SHUTDOWN						
T_{SDN}	Thermal shutdown threshold	Shutdown temperature (2)		165		°C
		Hysteresis (2)		30		
ON-TIME TIMER CONTROL						
T_{ON}	On time	V_{IN} = 12 V, V_O = 1.05 V		145		ns
$T_{OFF(MIN)}$	Minimum off time	T_A = 25°C, V_{FB} = 0.7 V		260	310	ns
SOFT START						
I_{SSC}	SS charge current	V_{SS} = 0 V	1.4	2.0	2.6	µA
I_{SSD}	SS discharge current	V_{SS} = 0.5 V	0.1	0.2		mA
POWER GOOD						
V_{THPG}	PG threshold	V_{FB} rising (good)	85	90	95	%
		V_{FB} falling (fault)		85		%
I_{PG}	PG sink current	PG = 0.5 V	2.5	5		mA
OUTPUT UNDERVOLTAGE AND OVERVOLTAGE PROTECTION						
V_{OVP}	Output OVP trip threshold	OVP detect	115	120	125	%
T_{OVPDEL}	Output OVP prop delay			10		µs
V_{UVP}	Output UVP trip threshold	UVP detect	60	65	70	%
		Hysteresis		10		%
T_{UVPDEL}	Output UVP delay			0.25		ms
T_{UVPEN}	Output UVP enable delay	Relative to soft-start time		x 1.7		
UVLO						
V_{UVLO}	UVLO threshold	Wake up V_{REG5} voltage	3.5	3.8	4.1	V
		Hysteresis V_{REG5} voltage	0.23	0.35	0.47	

(2) Not production tested.

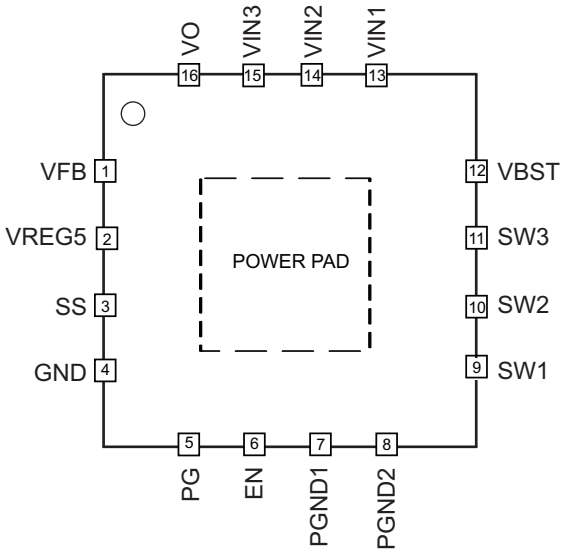
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DEVICE INFORMATION

PWP PACKAGE
(TOP VIEW)



RSA PACKAGE
(TOP VIEW)

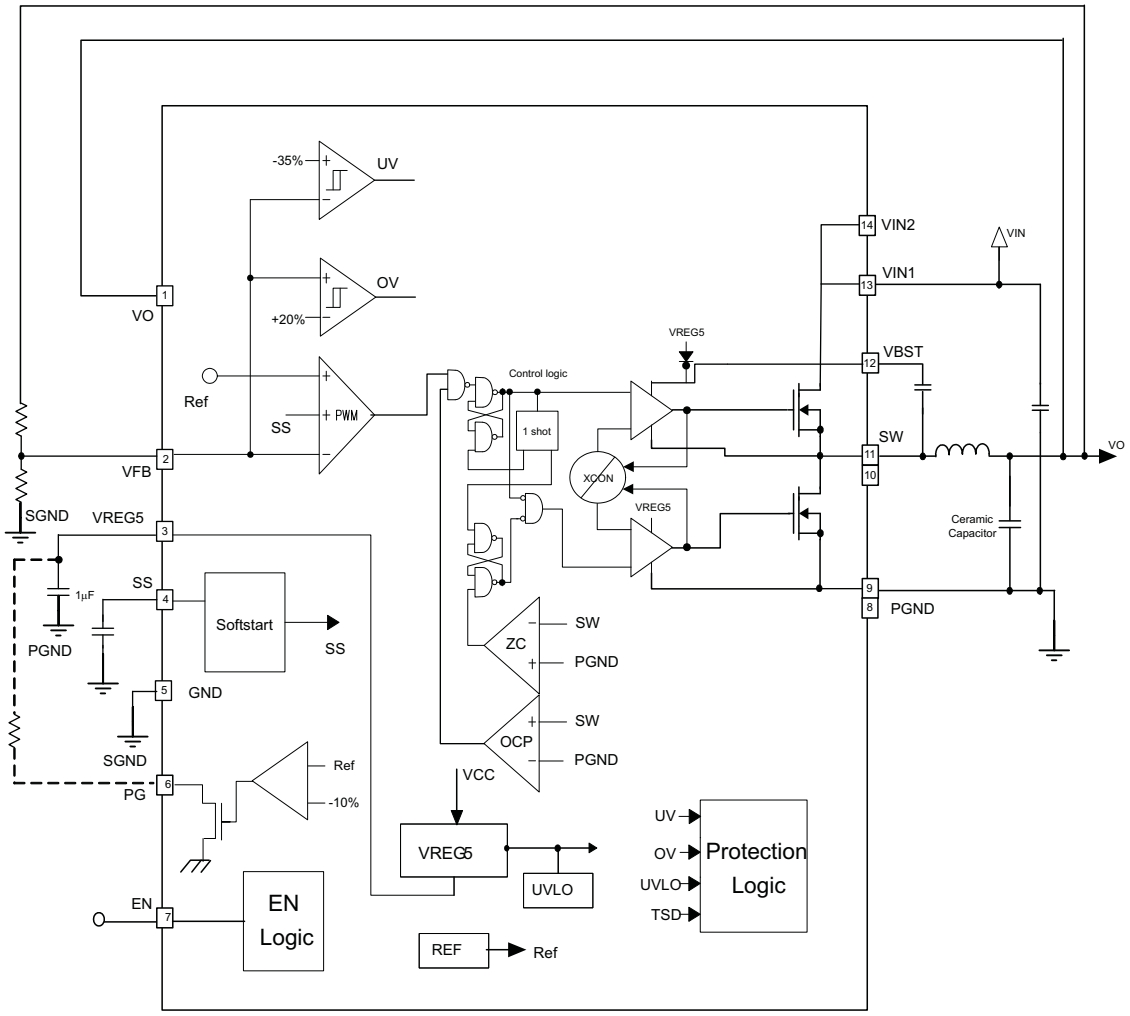


PIN FUNCTIONS

PIN			DESCRIPTION
NAME	NUMBER		
	PWP 14	RSA 16	
VO	1	16	Connect to output of converter. This terminal is used for On-Time Adjustment.
VFB	2	1	Converter feedback input. Connect to output voltage with feedback resistor divider.
VREG5	3	2	5.5 V power supply output. A capacitor (typical 1 µF) should be connected to GND. VREG5 is not active when EN is low.
SS	4	3	Soft-start control. A external capacitor should be connected to GND.
GND	5	4	Signal ground pin.
PG	6	5	Open drain power good output.
EN	7	6	Enable control input. EN is active high and must be pulled up to enable the device.
PGND1, PGND2	8, 9	7, 8	Ground returns for low-side MOSFET. Also serve as inputs of current comparators. Connect PGND and GND strongly together near the IC.
SW1, SW2	10, 11	9, 10, 11	Switch node connection between high-side NFET and low-side NFET. Also serve as inputs to current comparators.
VBST	12	12	Supply input for high-side NFET gate driver (boost terminal). Connect capacitor from this pin to respective SW1, SW2 terminals. An internal PN diode is connected between VREG5 to VBST pin.
VIN1, VIN2	13, 14	13, 14, 15	Power input and connected to high side NFET drain. Supply input for 5-V internal linear regulator for the control circuitry.
PowerPAD™	Back side	Back side	Thermal pad of the package. Must be soldered to achieve appropriate dissipation. Should be connected to PGND.

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FUNCTIONAL BLOCK DIAGRAM



A. The block diagram shown is for the PWP 14 pin package. The QFN 16 pin package block diagram is identical except for the pin out.

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OVERVIEW

The TPS54426 is a 4-A synchronous step-down (buck) converter with two integrated N-channel MOSFETs and auto-skip Eco-mode™ to improve light lode efficiency. It operates using D-CAP2™ mode control. The fast transient response of D-CAP2™ control reduces the output capacitance required to meet a specific level of performance. Proprietary internal circuitry allows the use of low ESR output capacitors including ceramic and special polymer types.

DETAILED DESCRIPTION

PWM Operation

The main control loop of the TPS54426 is an adaptive on-time pulse width modulation (PWM) controller that supports a proprietary D-CAP2™ mode control. D-CAP2™ mode control combines constant on-time control with an internal compensation circuit for pseudo-fixed frequency and low external component count configuration with both low ESR and ceramic output capacitors. It is stable with virtually no ripple at the output.

At the beginning of each cycle, the high-side MOSFET is turned on. The MOSFET is turned off after the internal one-shot timer expires. The one-shot timer is set by the converter input voltage, VIN, and the output voltage, VO, to maintain a pseudo-fixed frequency over the input voltage range, hence it is called adaptive on-time control. The one-shot timer is reset and the high-side MOSFET is turned on again when the feedback voltage falls below the reference voltage. An internal ramp is added to reference voltage to simulate output ripple, eliminating the need for ESR induced output ripple from D-CAP2™ mode control.

PWM Frequency and Adaptive On-Time Control

TPS54426 uses an adaptive on-time control scheme and does not have a dedicated on board oscillator. The TPS54426 runs with a pseudo-constant frequency of 700 kHz by using the input voltage and output voltage to set the on-time one-shot timer. The on-time is inversely proportional to the input voltage and proportional to the output voltage, therefore, when the duty ratio is VOUT/VIN, the frequency is constant.

Auto-Skip Eco-Mode™ Control

The TPS54426 is designed with Auto-Skip Eco-mode™ to increase light load efficiency. As the output current decreases from heavy load condition, the inductor current is also reduced and eventually comes to point that its rippled valley touches zero level, which is the boundary between continuous conduction and discontinuous conduction modes. The rectifying MOSFET is turned off when its zero inductor current is detected. As the load current further decreases the converter run into discontinuous conduction mode. The on-time is kept almost the same as is was in the continuous conduction mode so that it takes longer time to discharge the output capacitor with smaller load current to the level of the reference voltage. The transition point to the light load operation $I_{OUT(LL)}$ current can be calculated in Equation 1.

$$I_{OUT(LL)} = \frac{1}{2 \cdot L \cdot fsw} = \frac{(V_{IN} - V_{OUT}) \cdot V_{OUT}}{V_{IN}} \quad (1)$$

Soft Start and Pre-Biased Soft Start

The soft start function is adjustable. When the EN pin becomes high, 2 µA current begins charging the capacitor which is connected from the SS pin to GND. Smooth control of the output voltage is maintained during start up. The equation for the slow start time is shown in Equation 2. VFB voltage is 0.765 V and SS pin source current is 2 µA.

$$Tss(ms) = \frac{C6(nF) \bullet Vref}{Iss(\mu A)} = \frac{C6(nF) \bullet 0.765}{2} \quad (2)$$

The TPS54426 contains a unique circuit to prevent current from being pulled from the output during startup if the output is pre-biased. When the soft-start commands a voltage higher than the pre-bias level (internal soft start becomes greater than feedback voltage V_{FB}), the controller slowly activates synchronous rectification by starting the first low side FET gate driver pulses with a narrow on-time. It then increments that on-time on a cycle-by-cycle basis until it coincides with the time dictated by (1-D), where D is the duty cycle of the converter. This scheme prevents the initial sinking of the pre-bias output, and ensure that the out voltage (VO) starts and ramps up smoothly into regulation and the control loop is given time to transition from pre-biased start-up to normal mode operation.

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Power Good

The TPS54426 has power-good open drain output. The power good function is activated after soft start has finished. The power good function becomes active after 1.7 times soft-start time. When the output voltage is within -10% of the target value, internal comparators detect power good state and the power good signal becomes high. Rpg resister value ,which is connected between PG and VREG5, is required from 20kΩ to 150kΩ. If the feedback voltage goes under 15% of the target value, the power good signal becomes low after a 5 µs internal delay.

VREG5

VREG5 is an internally generated voltage source used by the TPS54425. It is derived directly from the input voltage and is nominally regulated to 5.5 V when the input voltage is above 5.6 V. The output of the VREG5 regulator is the input to the internal UVLO function. VREG5 must be above the UVLO wake up threshold voltage (3.8 V typical) for the TPS54425 to function. Connect a 1.0 µF capacitor between pin 3 of the TPS54425 and power ground for proper regulation of the VREG5 output. The VREG5 output voltage is available for external use and can typically source up to 70 mA. The VREG5 output is disabled when the TPS54425 EN pin is open or pulled low.

Output Discharge Control

TPS54426 discharges the output when EN is low, or the controller is turned off by the protection functions (OVP, UVP, UVLO and thermal shutdown). The output is discharged by an internal 50-Ω MOSFET which is connected from VO to PGND. The internal low-side MOSFET is not turned on during the output discharge operation to avoid the possibility of causing negative voltage at the output.

Current Protection

The output overcurrent protection (OCP) is implemented using a cycle-by-cycle valley detect control circuit. The switch current is monitored by measuring the low-side FET switch voltage between the SW pin and GND. This voltage is proportional to the switch current. To improve accuracy, the voltage sensing is temperature compensated.

During the on-time of the high-side FET switch, the switch current increases at a linear rate determined by V_{IN}, V_{OUT}, the on-time, and the output inductor value. During the on-time of the low-side FET switch, this current decreases linearly. The average value of the switch current is the load current I_{OUT}. If the measured voltage is above the voltage proportional to the current limit. Then, the device constantly monitors the low-side FET switch voltage, which is proportional to the switch current, during the low-side on-time.

The converter maintains the low-side switch on until the measured voltage is below the voltage corresponding to the current limit at which time the switching cycle is terminated and a new switching cycle begins. In subsequent switching cycles, the on-time is set to a fixed value and the current is monitored in the same manner.

There are some important considerations for this type of overcurrent protection. The load current one half of the peak-to-peak inductor current higher than the overcurrent threshold. Also when the current is being limited, the output voltage tends to fall as the demanded load current may be higher than the current available from the converter. This may cause the output under-voltage protection circuit to be activated. When the overcurrent condition is removed, the output voltage will return to the regulated value. This protection is non-latching.

Over/Under Voltage Protection

TPS54426 monitors a resistor divided feedback voltage to detect over and under voltage. When the feedback voltage becomes higher than 120% of the target voltage, the OVP comparator output goes high and the circuit latches as the high-side MOSFET driver turns off and the low-side MOSFET turns on. When the feedback voltage becomes lower than 65% of the target voltage, the UVP comparator output goes high and an internal UVP delay counter begins. After 250 µs, the device latches off both internal top and bottom MOSFET. This function is enabled approximately 1.7 x softstart time.

UVLO Protection

Undervoltage lock out protection (UVLO) monitors the voltage of the V_{REG5} pin. When the V_{REG5} voltage is lower than UVLO threshold voltage, the TPS54426 is shut off. This is protection is non-latching.

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Thermal Shutdown

TPS54426 monitors the temperature of itself. If the temperature exceeds the threshold value (typically 165°C), the device is shut off. This is non-latch protection.

TYPICAL CHARACTERISTICS

Vin CURRENT vs JUNCTION TEMPERATURE

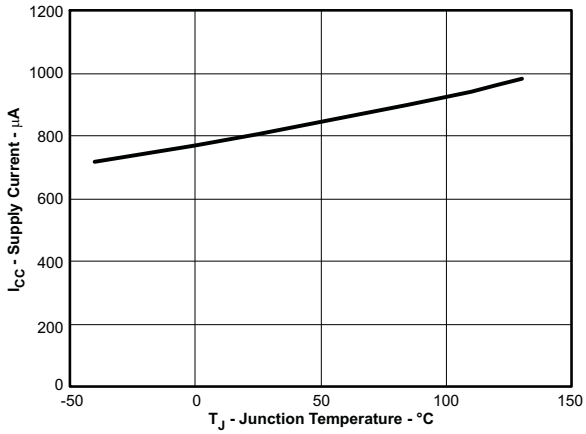


Figure 1.

Vin SHUTDOWN CURRENT vs JUNCTION TEMPERATURE

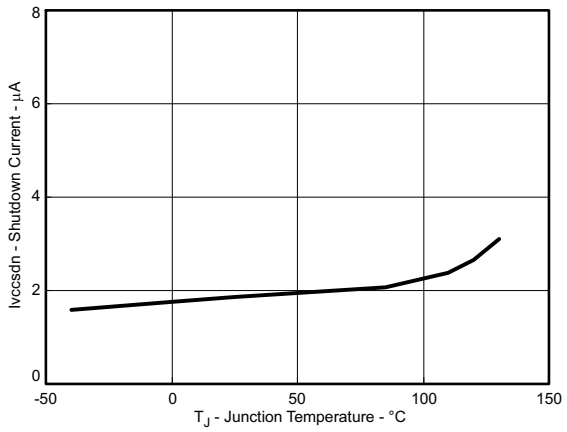


Figure 2.

EN CURRENT vs EN VOLTAGE

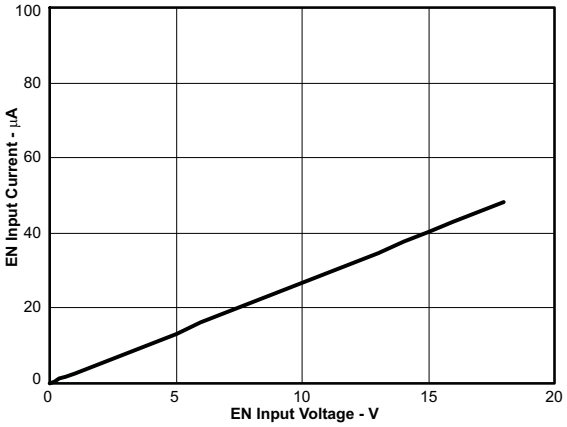


Figure 3.

1.05V OUTPUT VOLTAGE vs OUTPUT CURRENT

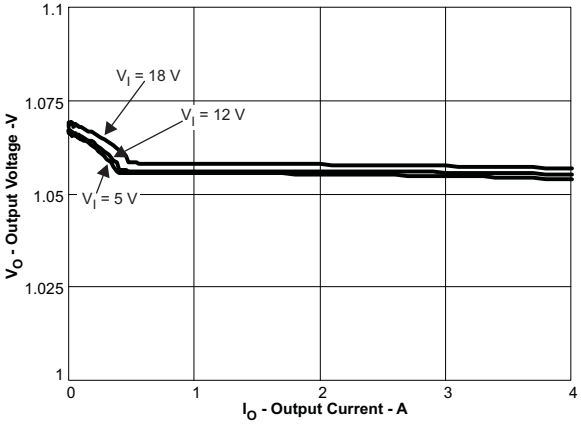


Figure 4.

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TYPICAL CHARACTERISTICS (continued)

1.05V OUTPUT VOLTAGE vs INPUT VOLTAGE

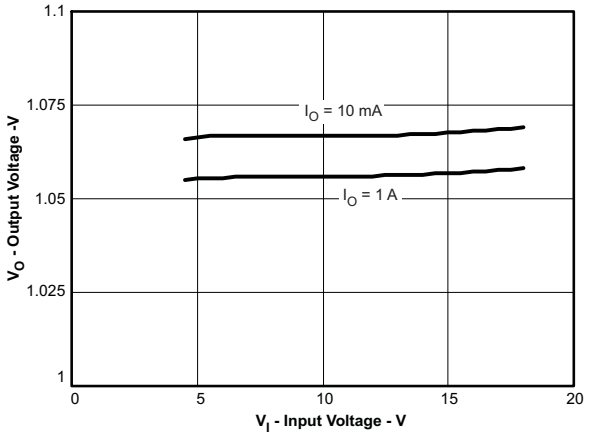


Figure 5.

1.05V 50mA to 4A LOAD TRANSIENT RESPONSE

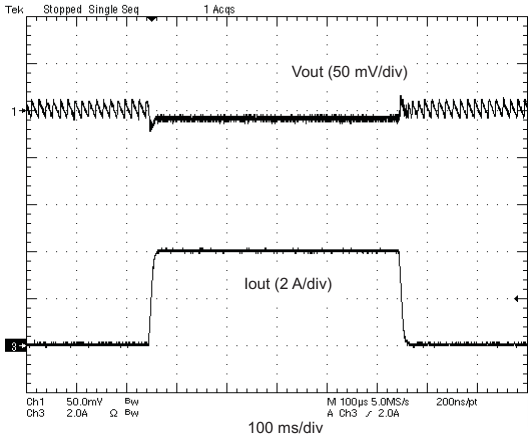


Figure 6.

STARTUP WAVEFORM

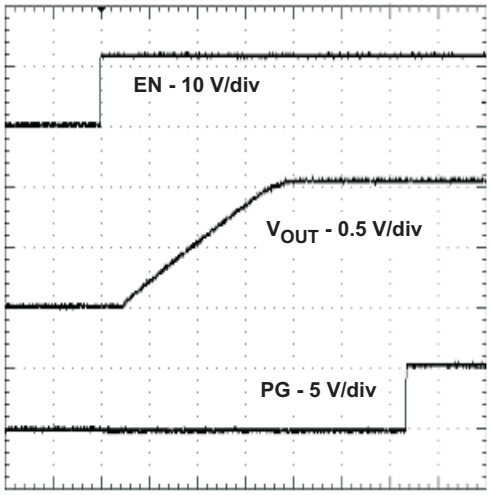


Figure 7.

EFFICIENCY vs OUTPUT CURRENT

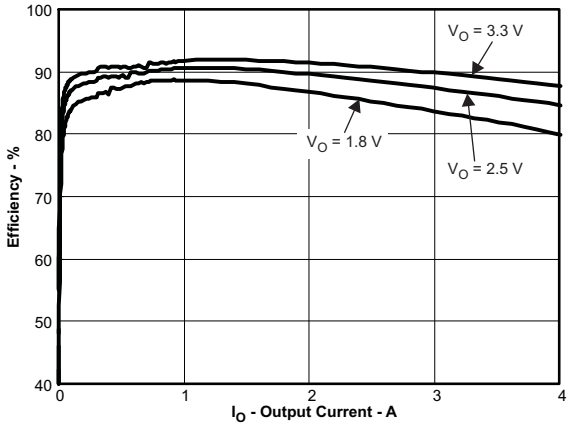


Figure 8.

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TYPICAL CHARACTERISTICS (continued)

LIGHT LOAD EFFICIENCY vs OUTPUT CURRENT

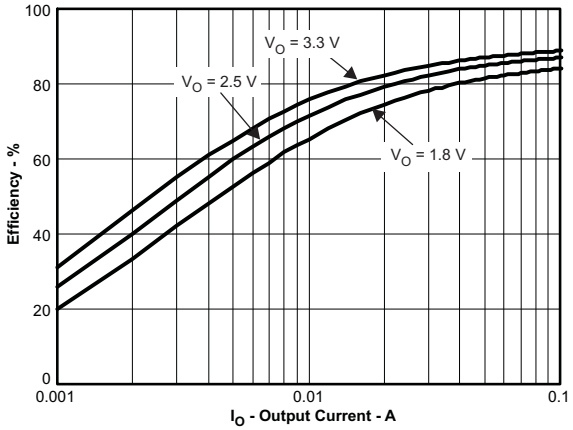


Figure 9.

SWITCHING FREQUENCY vs INPUT VOLTAGE (IO=1A)

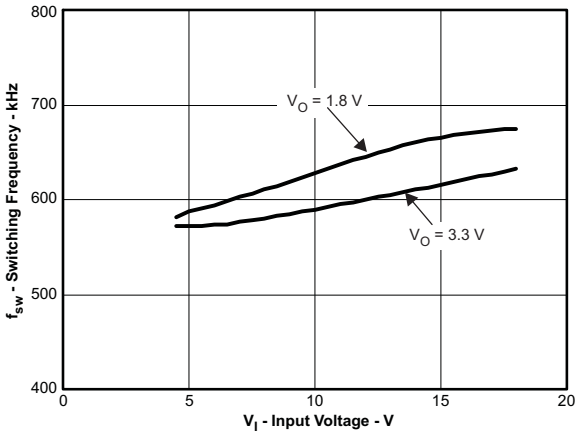


Figure 10.

WITCHING FREQUENCY vs OUTPUT CURRENT

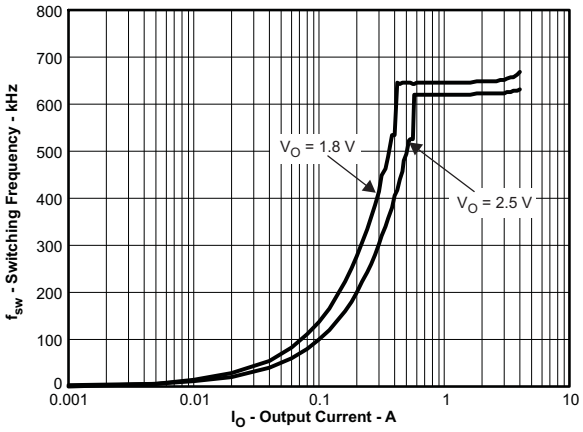


Figure 11.

VOLTAGE RIPPLE AT OUTPUT (IO=4A)

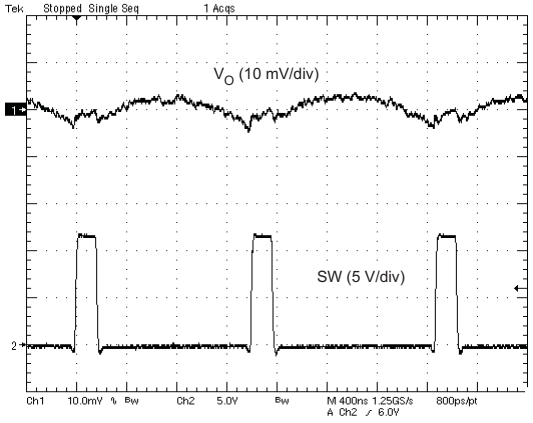


Figure 12.

VOLTAGE RIPPLE AT INPUT (IO=4A)

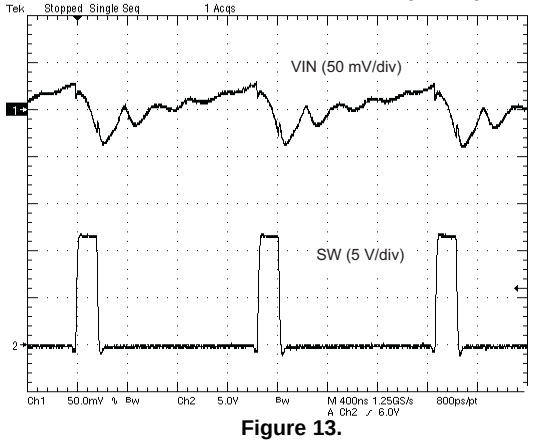


Figure 13.

DESIGN GUIDE

Step By Step Design Procedure

To begin the design process, you must know a few application parameters:

- Input voltage range
- Output voltage
- Output current
- Output voltage ripple
- Input voltage ripple

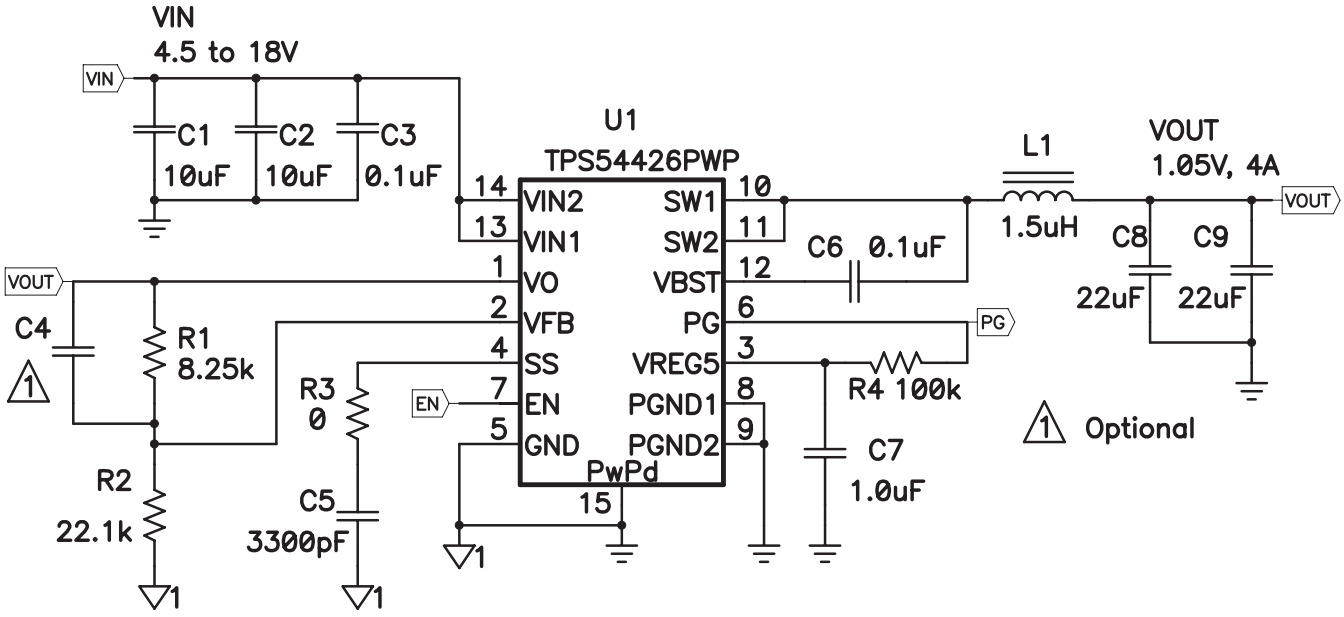


Figure 14. Schematic Diagram for This Design Example

Output Voltage Resistors Selection

The output voltage is set with a resistor divider from the output node to the VFB pin. It is recommended to use 1% tolerance or better divider resistors. Start by using Equation 3 and Equation 4 to calculate V_{OUT}

To improve efficiency at very light loads consider using larger value resistors, too high of resistance will be more susceptible to noise and voltage errors from the VFB input current will be more noticeable

For output voltage from 0.76 V to 2.5 V:

$$V_{OUT} = 0.765 \bullet \left(1 + \frac{R1}{R2}\right) \quad (3)$$

For output voltage over 2.5 V:

$$V_{OUT} = (0.763 + 0.0017 \bullet V_{OUT_SET}) \bullet \left(1 + \frac{R1}{R2}\right) \quad (4)$$

Where:

V_{OUT_SET} = Target V_{OUT} voltage

Output Filter Selection

The output filter used with the TPS54426 is an LC circuit. This LC filter has double pole at:

$$F_P = \frac{1}{2\pi\sqrt{L_{OUT} \times C_{OUT}}} \quad (5)$$

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At low frequencies, the overall loop gain is set by the output set-point resistor divider network and the internal gain of the TPS54426. The low frequency phase is 180 degrees. At the output filter pole frequency, the gain rolls off at a -40 dB per decade rate and the phase drops rapidly. D-CAP2™ introduces a high frequency zero that reduces the gain roll off to -20 dB per decade and increases the phase to 90 degrees one decade above the zero frequency. The inductor and capacitor selected for the output filter must be selected so that the double pole of Equation 5 is located below the high frequency zero but close enough that the phase boost provided be the high frequency zero provides adequate phase margin for a stable circuit. To meet this requirement use the values recommended in Table 1

Table 1. Recommended Component Values

Output Voltage (V)	R1 (kΩ)	R2 (kΩ)	C4 (pF)(1)	L1 (µH)	C8 + C9 (µF)
1	6.81	22.1		1.5	22 - 68
1.05	8.25	22.1		1.5	22 - 68
1.2	12.7	22.1		1.5	22 - 68
1.8	30.1	22.1	10 - 22	2.2	22 - 68
2.5	49.9	22.1	10 - 22	2.2	22 - 68
3.3	73.2	22.1	10 - 22	2.2	22 - 68
5	121	22.1	10 - 22	3.3	22 - 68

(1) Optional

For higher output voltages at or above 1.8 V, additional phase boost can be achieved by adding a feed forward capacitor (C4) in parallel with R1.

Since the DC gain is dependent on the output voltage, the required inductor value will increase as the output voltage increases. For higher output voltages above 1.8 V, additional phase boost can be achieved by adding a feed forward capacitor (C4) in parallel with R1

The inductor peak-to-peak ripple current, peak current and RMS current are calculated using Equation 6, Equation 7 and Equation 8. The inductor saturation current rating must be greater than the calculated peak current and the RMS or heating current rating must be greater than the calculated RMS current. Use 700 kHz for f_{SW}.

$$Ilp\text{-}p = \frac{V_{OUT}}{V_{IN\,(max)}} \bullet \frac{V_{IN\,(max)} - V_{OUT}}{L_O \bullet f_{SW}} \tag{6}$$

$$I_{lpeak} = I_O + \frac{Ilp\text{-}p}{2} \tag{7}$$

$$I_{Lo(RMS)} = \sqrt{I_O^2 + \frac{1}{12} Ilp\text{-}p^2} \tag{8}$$

For this design example, the calculated peak current is 4.47A and the calculated RMS current is 4.009 A. The inductor used is a TDK SPM6530-1R5M100 with a peak current rating of 11.5 A and an RMS current rating of 11 A.

The capacitor value and ESR determines the amount of output voltage ripple. The TPS54426 is intended for use with ceramic or other low ESR capacitors. Recommended values range from 22uF to 68uF. Use Equation 9 to determine the required RMS current rating for the output capacitor

$$I_{CO(RMS)} = \frac{V_{OUT} \bullet (V_{IN} - V_{OUT})}{\sqrt{12} \bullet V_{IN} \bullet L_O \bullet f_{SW}} \tag{9}$$

For this design two TDK C3216X5R0J226M 22uF output capacitors are used. The typical ESR is 2 mΩ each. The calculated RMS current is .271A and each output capacitor is rated for 4A.

Input Capacitor Selection

The TPS54426 requires an input decoupling capacitor and a bulk capacitor is needed depending on the application. A ceramic capacitor over 10 uF. is recommended for the decoupling capacitor. An additional 0.1 µF capacitor from pin 14 to ground is recommended to improve the stability of the over-current limit function. The capacitor voltage rating needs to be greater than the maximum input voltage.

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Bootstrap Capacitor Selection

A 0.1 µF ceramic capacitor must be connected between the VBST to SW pin for proper operation. It is recommended to use a ceramic capacitor.

VREG5 Capacitor Selection

A 1.0 µF ceramic capacitor must be connected between the VREG5 to GND pin for proper operation. It is recommended to use a ceramic capacitor.

THERMAL INFORMATION

This PowerPad™ package incorporates an exposed thermal pad that is designed to be directly to an external heatsink. The thermal pad must be soldered directly to the printed board (PCB). After soldering, the PCB can be used as a heatsink. In addition, through the use of thermal vias, the thermal pad can be attached directly to the appropriate copper plane shown in the electrical schematic for the device, or alternatively, can be attached to a special heatsink structure designed into the PCB. This design optimizes the heat transfer from the integrated circuit (IC).

For additional information on the PowerPAD™ package and how to use the advantage of its heat dissipating abilities, refer to Technical Brief, PowerPAD™ Thermally Enhanced Package, Texas Instruments Literature No. SLMA002 and Application Brief, PowerPAD™ Made Easy, Texas Instruments Literature No. SLMA004.

The exposed thermal pad dimensions for this package are shown in the following illustration.

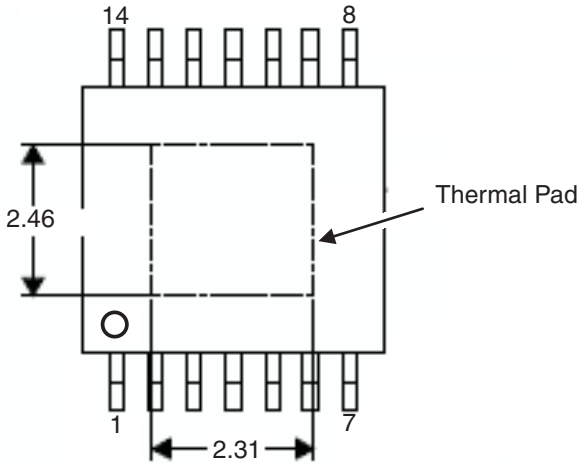


Figure 15. Thermal Pad Dimensions

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LAYOUT CONSIDERATIONS

1. Keep the input switching current loop as small as possible.
2. Keep the SW node as physically small and short as possible to minimize parasitic capacitance and inductance and to minimize radiated emissions. Kelvin connections should be brought from the output to the feedback pin of the device.
3. Keep analog and non-switching components away from switching components.
4. Make a single point connection from the signal ground to power ground.
5. Do not allow switching current to flow under the device.
6. Keep the pattern lines for VIN and PGND broad.
7. Exposed pad of device must be connected to PGND with solder.
8. VREG5 capacitor should be placed near the device, and connected PGND.
9. Output capacitor should be connected to a broad pattern of the PGND.
10. Voltage feedback loop should be as short as possible, and preferably with ground shield.
11. Lower resistor of the voltage divider which is connected to the VFB pin should be tied to SGND.
12. Providing sufficient via is preferable for VIN, SW and PGND connection.
13. PCB pattern for VIN, SW, and PGND should be as broad as possible.
14. VIN Capacitor should be placed as near as possible to the device.

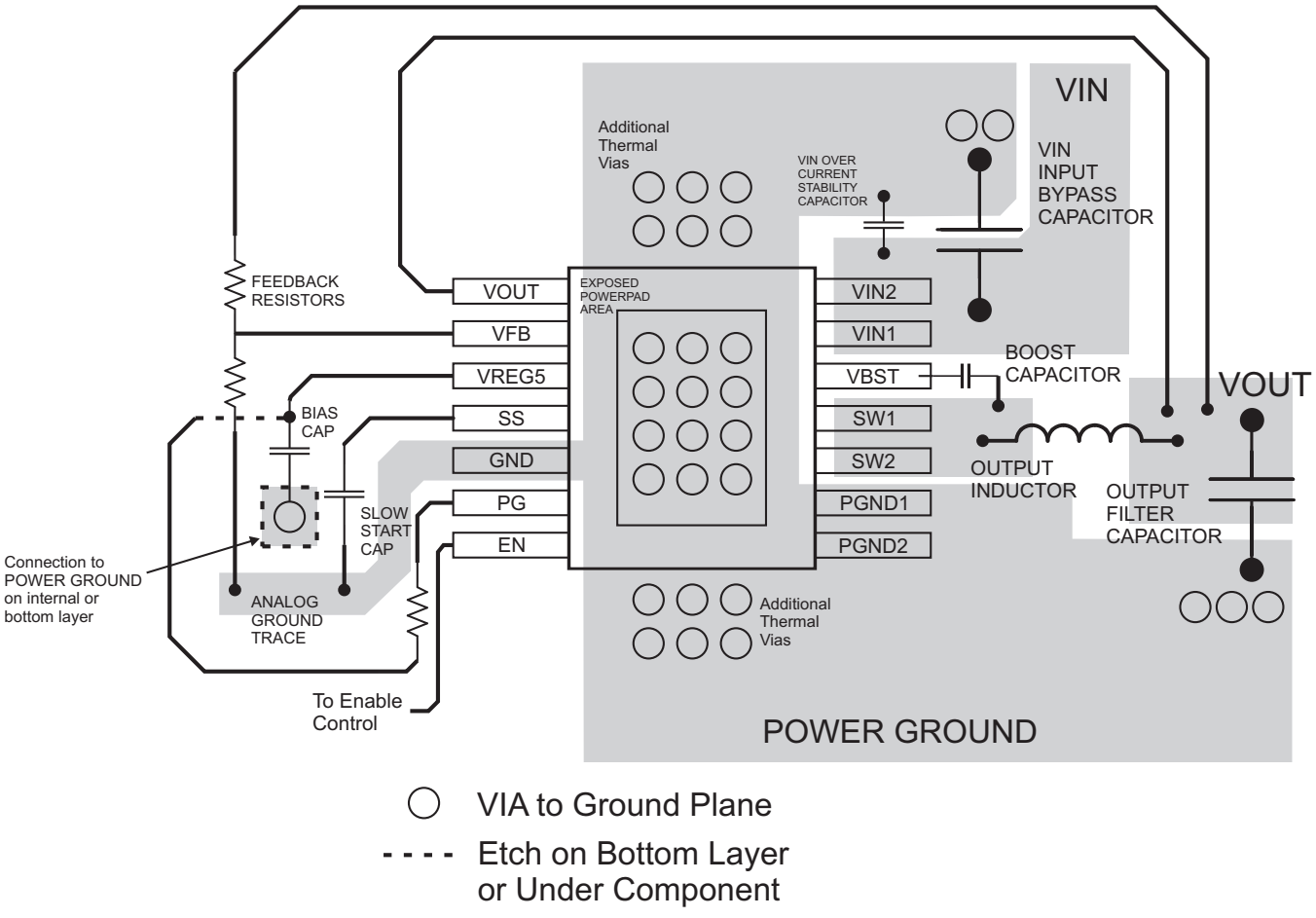


Figure 16. PCB Layout

REVISION HISTORY

Changes from Revision # (August 2010) to Revision A **Page**

- Added V_{IN} = 12V to condition in Electrical Characteristics table 3

Changes from Revision A (November 2010) to Revision B **Page**

- Added the RSA (16 pin QFN package) to: DESCRIPTION, ORDERING INFORMATION, THERMAL INFORMATION, and DEVICE INFORMATION sections 1
- Changed the Functional Block Diagram 6
- Changed the Power Good and Current Protection sections 8
- Added Note 1 to Table 1 13

Changes from Revision B (November 2010) to Revision C **Page**

- Changed ELEC CHAR table, UVLO threshold MIN from 3.55 V to 3.5 V, MAX from 4.05 V to 4.1 V 4

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PACKAGING INFORMATION

Orderable part number	Status (1)	Material type (2)	Package | Pins	Package qty | Carrier	RoHS (3)	Lead finish/ Ball material (4)	MSL rating/ Peak reflow (5)	Op temp (°C)	Part marking (6)
TPS54426PWP	Active	Production	HTSSOP (PWP) | 14	90 | TUBE	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 85	PS54426
TPS54426PWP.A	Active	Production	HTSSOP (PWP) | 14	90 | TUBE	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 85	PS54426
TPS54426PWP.B	Active	Production	HTSSOP (PWP) | 14	90 | TUBE	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 85	PS54426
TPS54426PWPR	Active	Production	HTSSOP (PWP) | 14	2000 | LARGE T&R	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 85	PS54426
TPS54426PWPR.A	Active	Production	HTSSOP (PWP) | 14	2000 | LARGE T&R	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 85	PS54426
TPS54426PWPR.B	Active	Production	HTSSOP (PWP) | 14	2000 | LARGE T&R	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 85	PS54426
TPS54426PWPRG4	Active	Production	HTSSOP (PWP) | 14	2000 | LARGE T&R	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 85	PS54426
TPS54426PWPRG4.A	Active	Production	HTSSOP (PWP) | 14	2000 | LARGE T&R	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 85	PS54426
TPS54426PWPRG4.B	Active	Production	HTSSOP (PWP) | 14	2000 | LARGE T&R	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 85	PS54426
TPS54426RSAR	Active	Production	QFN (RSA) | 16	3000 | LARGE T&R	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 85	TPS 54426
TPS54426RSAR.B	Active	Production	QFN (RSA) | 16	3000 | LARGE T&R	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 85	TPS 54426
TPS54426RSARG4	Active	Production	QFN (RSA) | 16	3000 | LARGE T&R	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 85	TPS 54426
TPS54426RSARG4.B	Active	Production	QFN (RSA) | 16	3000 | LARGE T&R	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 85	TPS 54426
TPS54426RSAT	Active	Production	QFN (RSA) | 16	250 | SMALL T&R	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 85	TPS 54426
TPS54426RSAT.B	Active	Production	QFN (RSA) | 16	250 | SMALL T&R	Yes	NIPDAU	Level-2-260C-1 YEAR	-40 to 85	TPS 54426

(1) **Status:** For more details on status, see our product life cycle.

(2) **Material type:** When designated, preproduction parts are prototypes/experimental devices, and are not yet approved or released for full production. Testing and final process, including without limitation quality assurance, reliability performance testing, and/or process qualification, may not yet be complete, and this item is subject to further changes or possible discontinuation. If available for ordering, purchases will be subject to an additional waiver at checkout, and are intended for early internal evaluation purposes only. These items are sold without warranties of any kind.

(3) **RoHS values:** Yes, No, RoHS Exempt. See the TI RoHS Statement for additional information and value definition.

(4) **Lead finish/Ball material:** Parts may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.

(5) **MSL rating/Peak reflow:** The moisture sensitivity level ratings and peak solder (reflow) temperatures. In the event that a part has multiple moisture sensitivity ratings, only the lowest level per JEDEC standards is shown. Refer to the shipping label for the actual reflow temperature that will be used to mount the part to the printed circuit board.

(6) **Part marking:** There may be an additional marking, which relates to the logo, the lot trace code information, or the environmental category of the part.

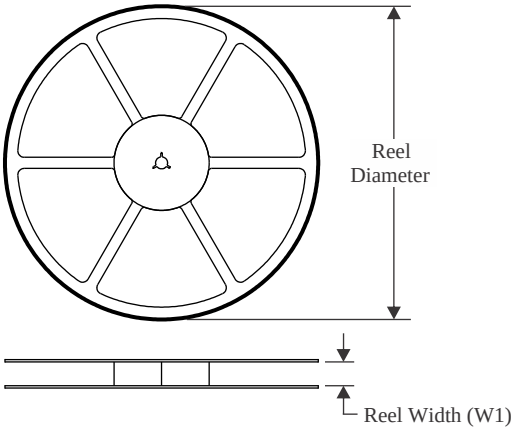
Multiple part markings will be inside parentheses. Only one part marking contained in parentheses and separated by a "~" will appear on a part. If a line is indented then it is a continuation of the previous line and the two combined represent the entire part marking for that device.

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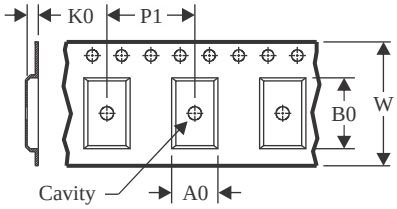
In no event shall TI's liability arising out of such information exceed the total purchase price of the TI part(s) at issue in this document sold by TI to Customer on an annual basis.

TAPE AND REEL INFORMATION

REEL DIMENSIONS

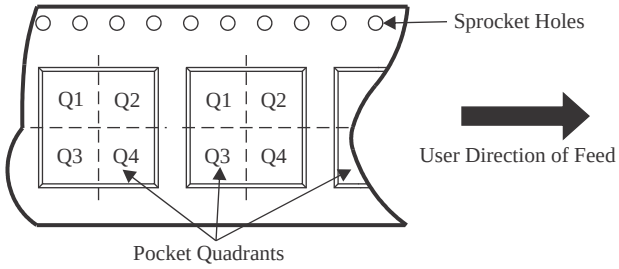


TAPE DIMENSIONS



A0	Dimension designed to accommodate the component width
B0	Dimension designed to accommodate the component length
K0	Dimension designed to accommodate the component thickness
W	Overall width of the carrier tape
P1	Pitch between successive cavity centers

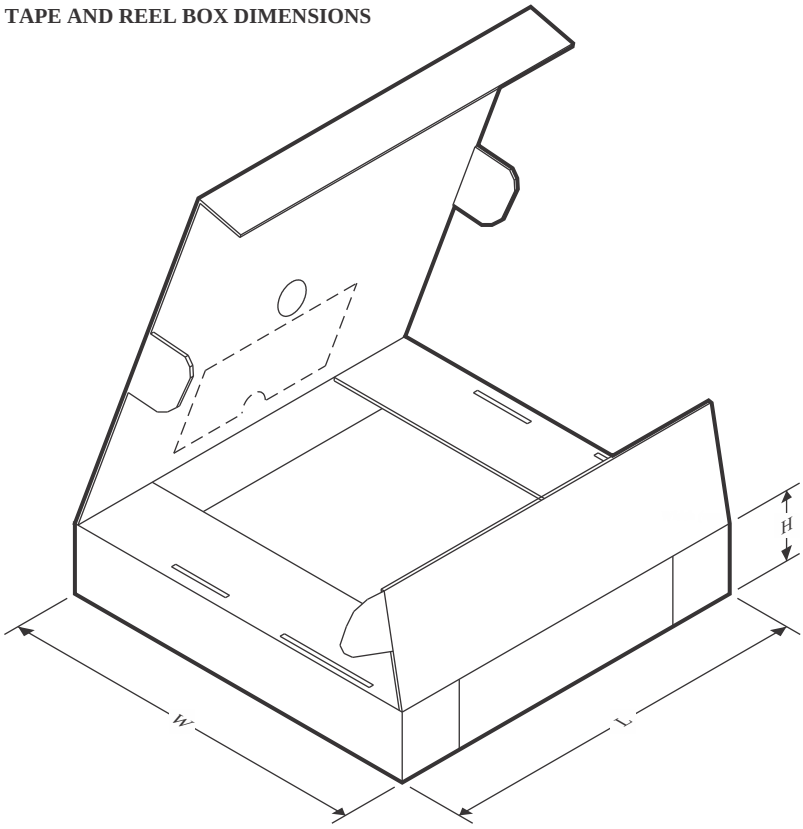
QUADRANT ASSIGNMENTS FOR PIN 1 ORIENTATION IN TAPE



*All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Reel Diameter (mm)	Reel Width W1 (mm)	A0 (mm)	B0 (mm)	K0 (mm)	P1 (mm)	W (mm)	Pin1 Quadrant
TPS54426PWPR	HTSSOP	PWP	14	2000	330.0	12.4	6.9	5.6	1.6	8.0	12.0	Q1
TPS54426PWPRG4	HTSSOP	PWP	14	2000	330.0	12.4	6.9	5.6	1.6	8.0	12.0	Q1
TPS54426RSAR	QFN	RSA	16	3000	330.0	12.4	4.25	4.25	1.15	8.0	12.0	Q2
TPS54426RSARG4	QFN	RSA	16	3000	330.0	12.4	4.25	4.25	1.15	8.0	12.0	Q2
TPS54426RSAT	QFN	RSA	16	250	180.0	12.4	4.25	4.25	1.15	8.0	12.0	Q2

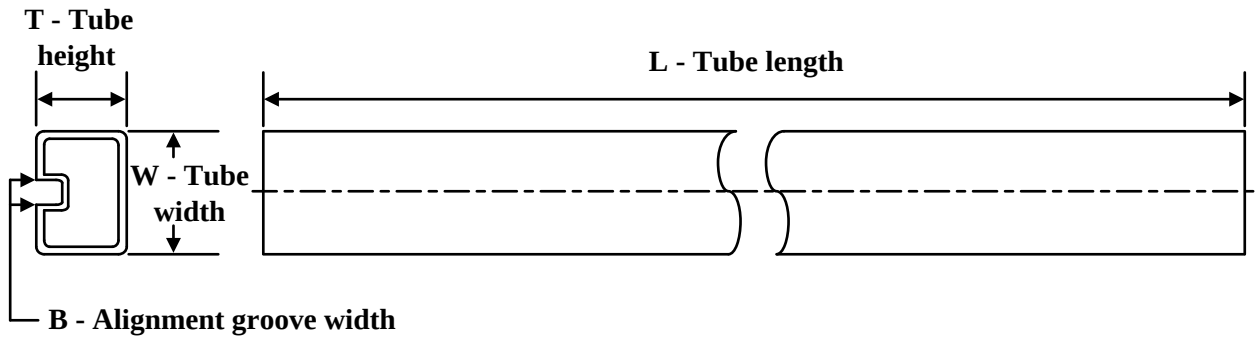
TAPE AND REEL BOX DIMENSIONS



*All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Length (mm)	Width (mm)	Height (mm)
TPS54426PWPR	HTSSOP	PWP	14	2000	353.0	353.0	32.0
TPS54426PWPRG4	HTSSOP	PWP	14	2000	350.0	350.0	43.0
TPS54426RSAR	QFN	RSA	16	3000	346.0	346.0	33.0
TPS54426RSARG4	QFN	RSA	16	3000	346.0	346.0	33.0
TPS54426RSAT	QFN	RSA	16	250	210.0	185.0	35.0

TUBE



*All dimensions are nominal

Device	Package Name	Package Type	Pins	SPQ	L (mm)	W (mm)	T (µm)	B (mm)
TPS54426PWP	PWP	HTSSOP	14	90	530	10.2	3600	3.5
TPS54426PWP	PWP	HTSSOP	14	90	530	10.2	3600	3.5
TPS54426PWP.A	PWP	HTSSOP	14	90	530	10.2	3600	3.5
TPS54426PWP.A	PWP	HTSSOP	14	90	530	10.2	3600	3.5
TPS54426PWP.B	PWP	HTSSOP	14	90	530	10.2	3600	3.5
TPS54426PWP.B	PWP	HTSSOP	14	90	530	10.2	3600	3.5

GENERIC PACKAGE VIEW

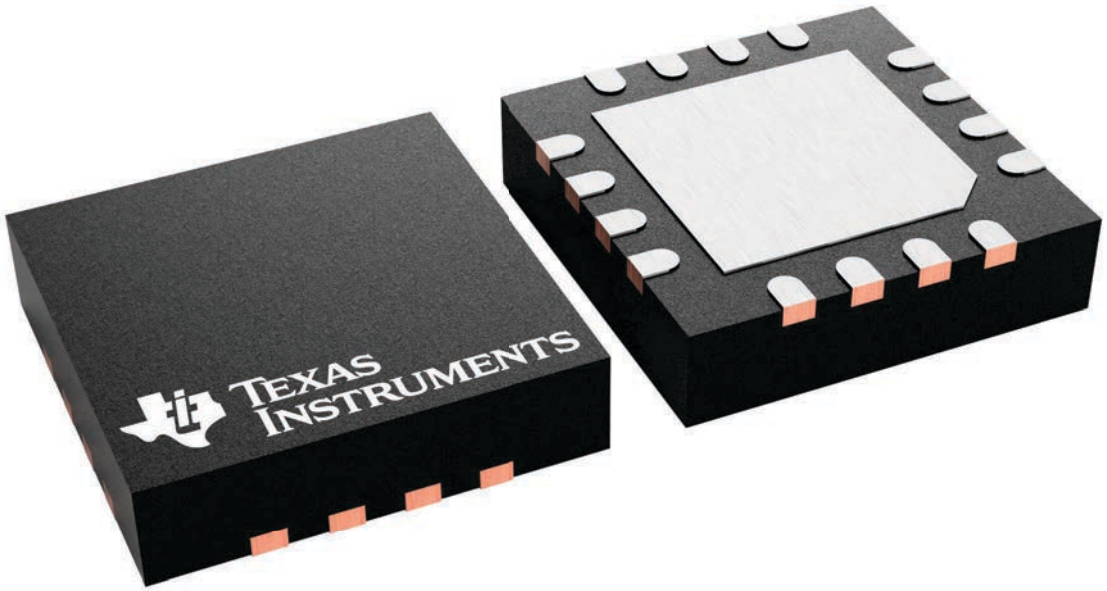
RSA 16

VQFN - 1 mm max height

4 x 4, 0.65 mm pitch

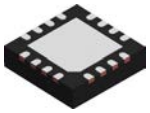
PLASTIC QUAD FLATPACK - NO LEAD

This image is a representation of the package family, actual package may vary.
Refer to the product data sheet for package details.



4230969/A

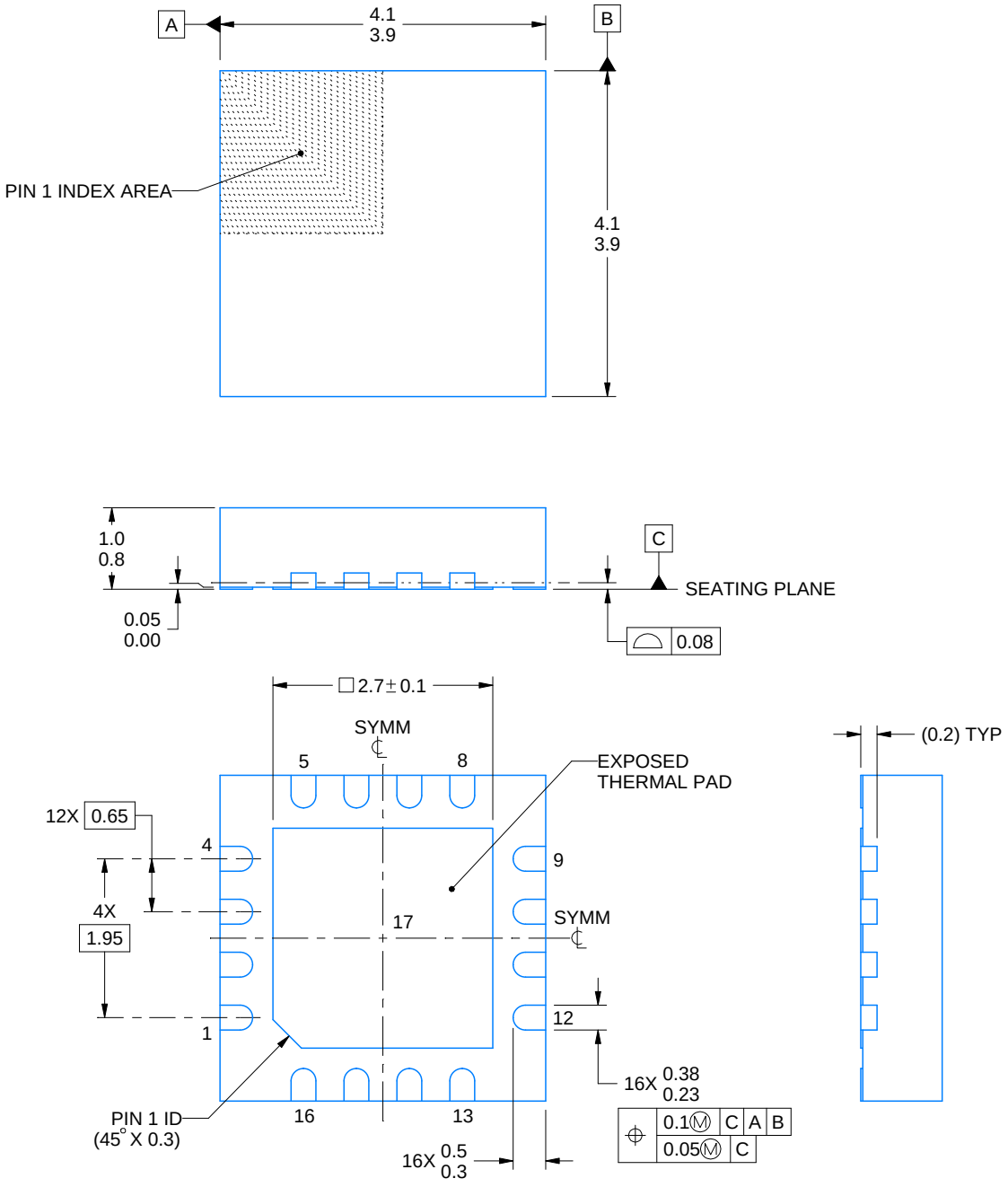
RSA0016B



PACKAGE OUTLINE

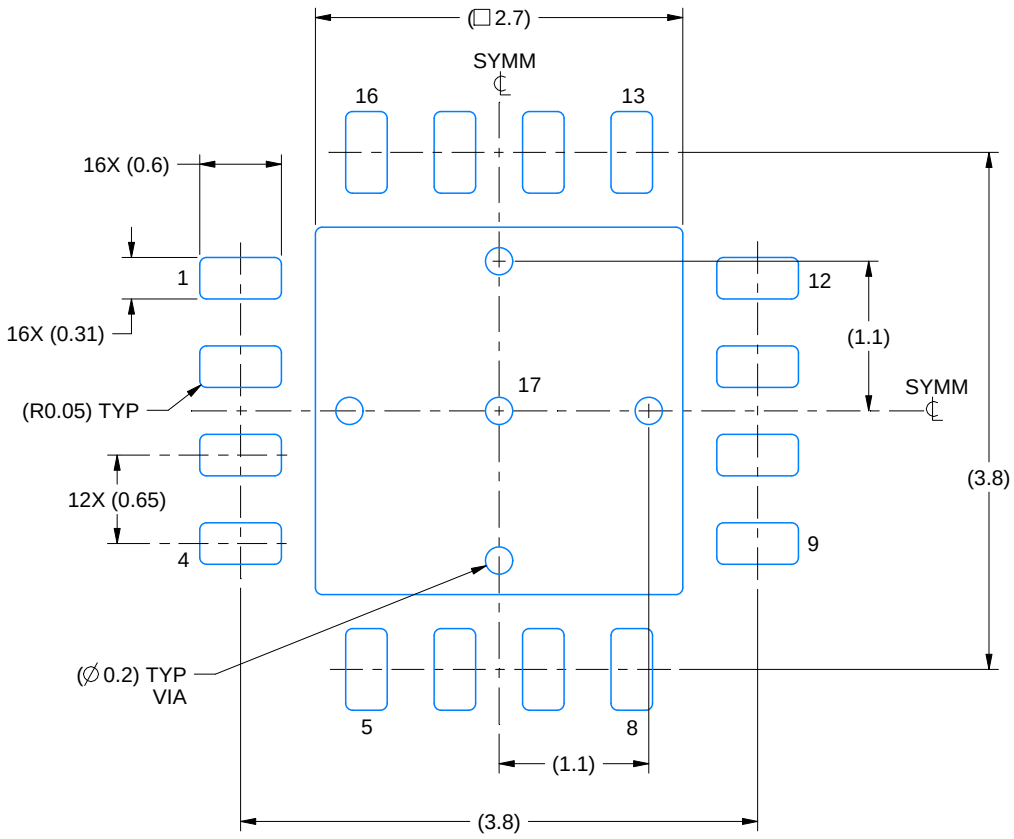
VQFN - 1 mm max height

PLASTIC QUAD FLATPACK - NO LEAD

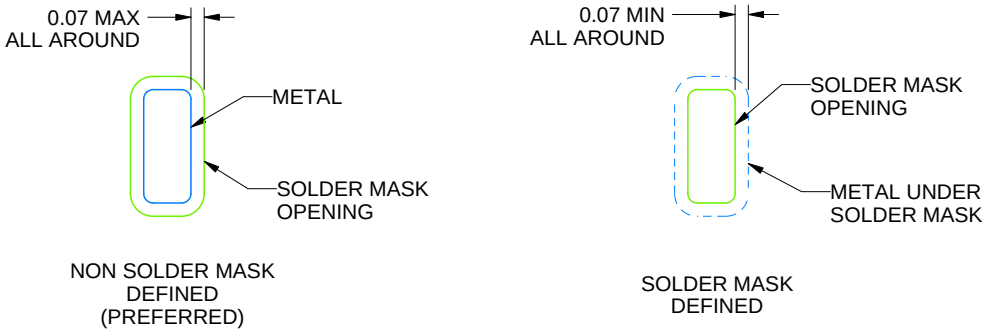


NOTES:

1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. The package thermal pad must be soldered to the printed circuit board for thermal and mechanical performance.
4. Reference JEDEC registration MO-220.



LAND PATTERN EXAMPLE
EXPOSED METAL SHOWN
SCALE:18X

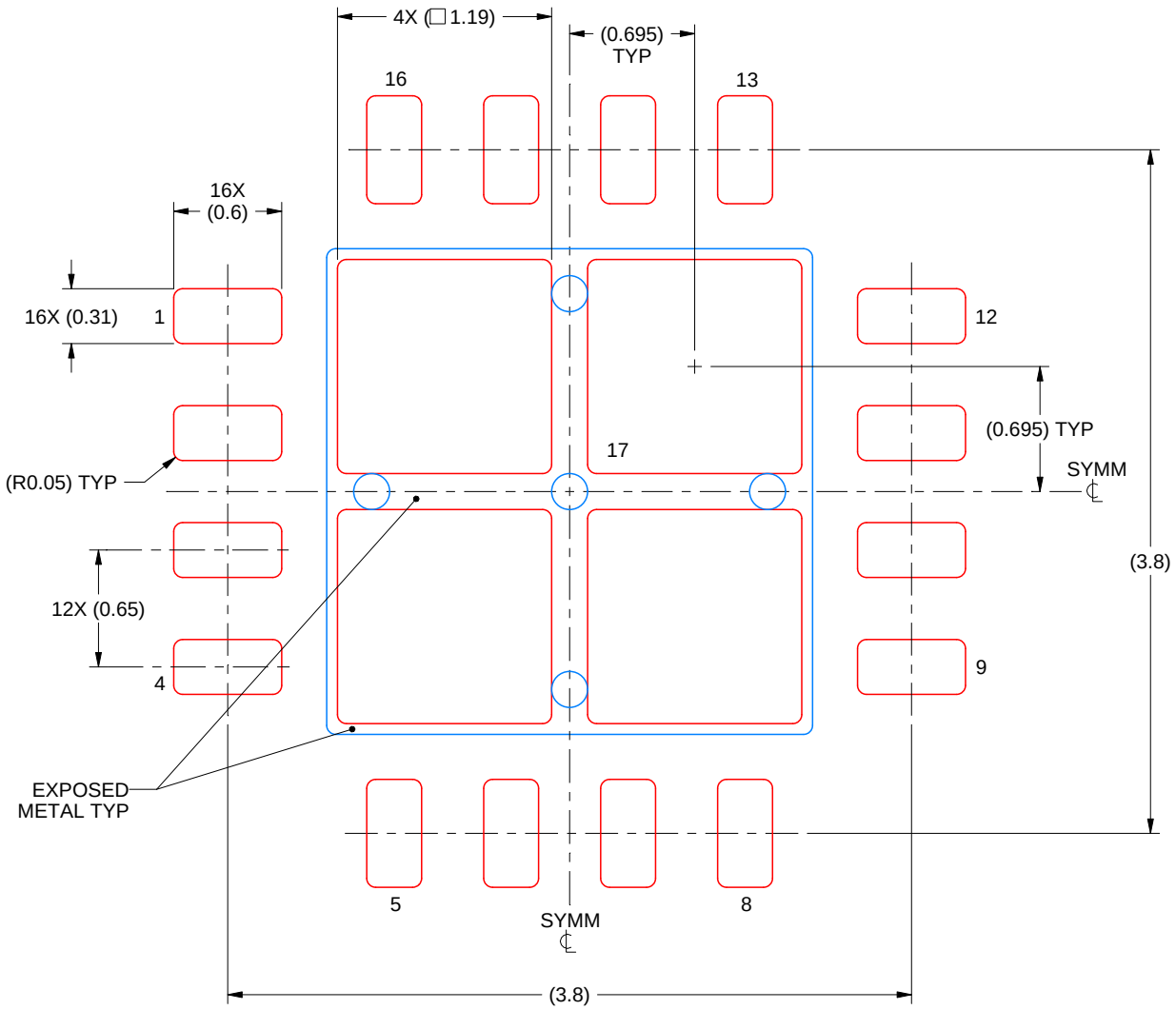


SOLDER MASK DETAILS

4219093/A 08/2021

NOTES: (continued)

5. This package is designed to be soldered to a thermal pad on the board. For more information, see Texas Instruments literature number SLUA271 (www.ti.com/lit/slua271).
6. Vias are optional depending on application, refer to device data sheet. If any vias are implemented, refer to their locations shown on this view. It is recommended that vias under paste be filled, plugged or tented.



SOLDER PASTE EXAMPLE
BASED ON 0.125 mm THICK STENCIL

EXPOSED PAD
77% PRINTED SOLDER COVERAGE BY AREA
SCALE:25X

4219093/A 08/2021

NOTES: (continued)

7. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.

GENERIC PACKAGE VIEW

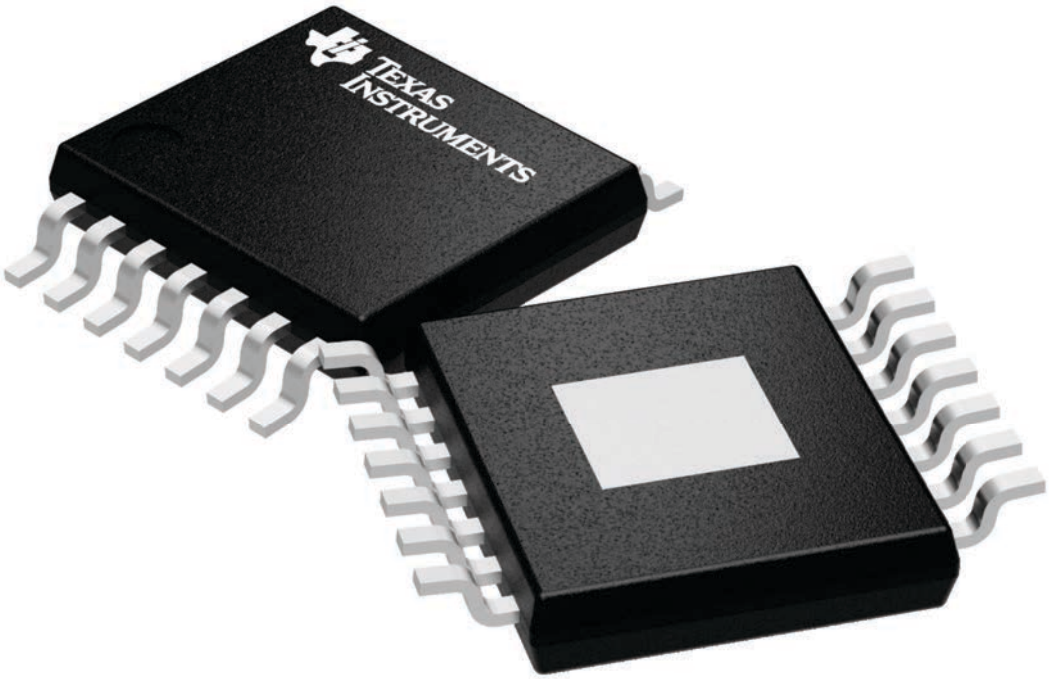
PWP 14

PowerPAD TSSOP - 1.2 mm max height

4.4 x 5.0, 0.65 mm pitch

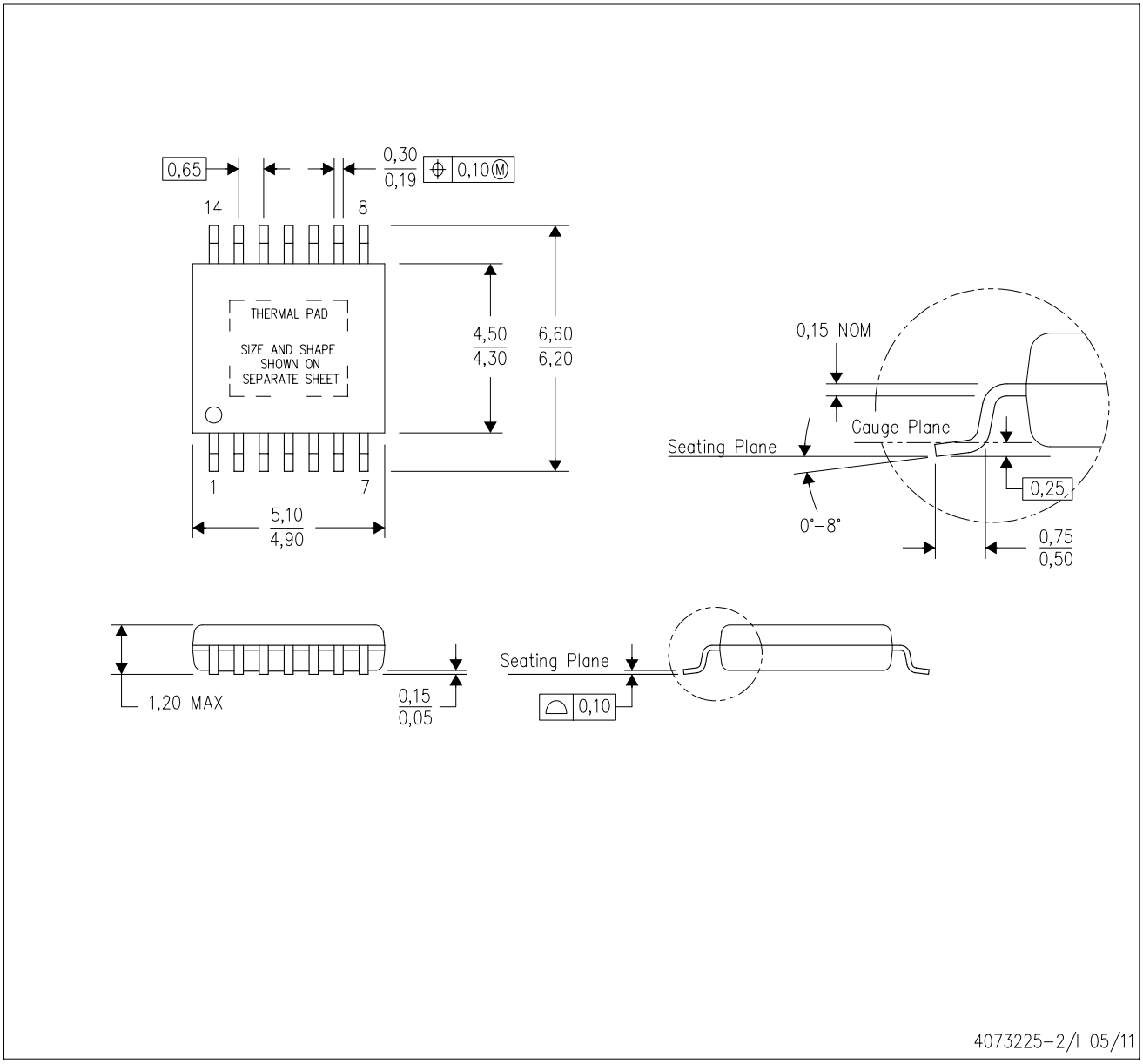
PLASTIC SMALL OUTLINE

This image is a representation of the package family, actual package may vary.
Refer to the product data sheet for package details.



4224995/A

PWP (R–PDSO–G14) PowerPAD™ PLASTIC SMALL OUTLINE



NOTES:
A. All linear dimensions are in millimeters.
B. This drawing is subject to change without notice.
C. Body dimensions do not include mold flash or protrusions. Mold flash and protrusion shall not exceed 0.15 per side.
D. This package is designed to be soldered to a thermal pad on the board. Refer to Technical Brief, PowerPad Thermally Enhanced Package, Texas Instruments Literature No. SLMA002 for information regarding recommended board layout. This document is available at www.ti.com <http://www.ti.com>.
E. See the additional figure in the Product Data Sheet for details regarding the exposed thermal pad features and dimensions.
E. Falls within JEDEC MO–153

PowerPAD is a trademark of Texas Instruments.

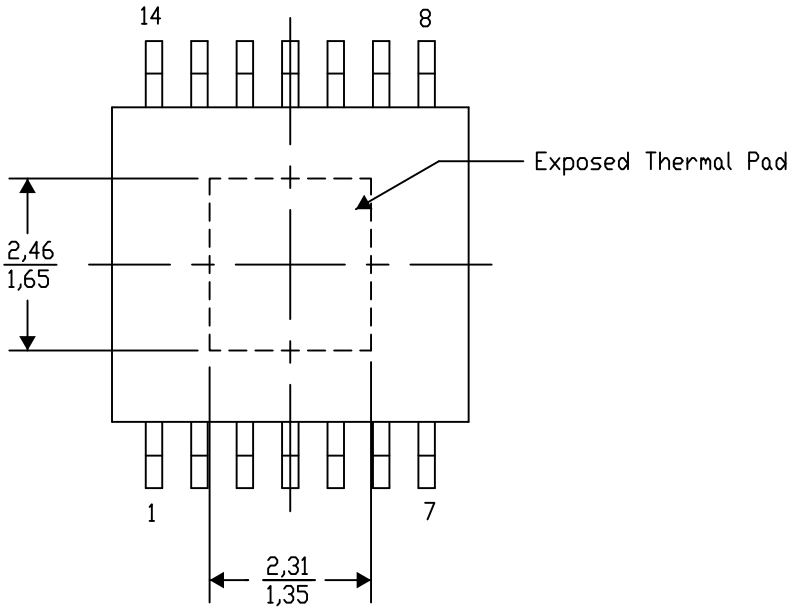
PWP (R–PDSO–G14) PowerPAD™ SMALL PLASTIC OUTLINE

THERMAL INFORMATION

This PowerPAD™ package incorporates an exposed thermal pad that is designed to be attached to a printed circuit board (PCB). The thermal pad must be soldered directly to the PCB. After soldering, the PCB can be used as a heatsink. In addition, through the use of thermal vias, the thermal pad can be attached directly to the appropriate copper plane shown in the electrical schematic for the device, or alternatively, can be attached to a special heatsink structure designed into the PCB. This design optimizes the heat transfer from the integrated circuit (IC).

For additional information on the PowerPAD package and how to take advantage of its heat dissipating abilities, refer to Technical Brief, PowerPAD Thermally Enhanced Package, Texas Instruments Literature No. SLMA002 and Application Brief, PowerPAD Made Easy, Texas Instruments Literature No. SLMA004. Both documents are available at www.ti.com.

The exposed thermal pad dimensions for this package are shown in the following illustration.



Top View

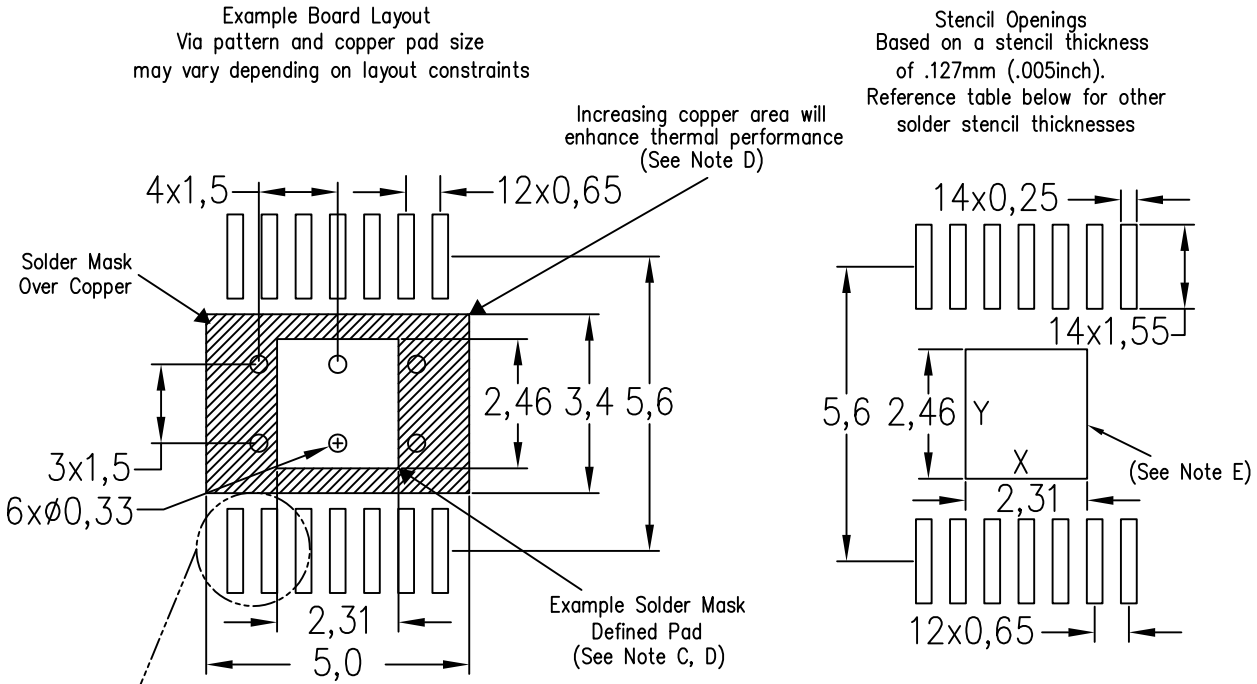
Exposed Thermal Pad Dimensions

4206332–2/AO 01/16

NOTE: A. All linear dimensions are in millimeters

PowerPAD is a trademark of Texas Instruments

PWP (R–PDSO–G14) PowerPAD™ PLASTIC SMALL OUTLINE

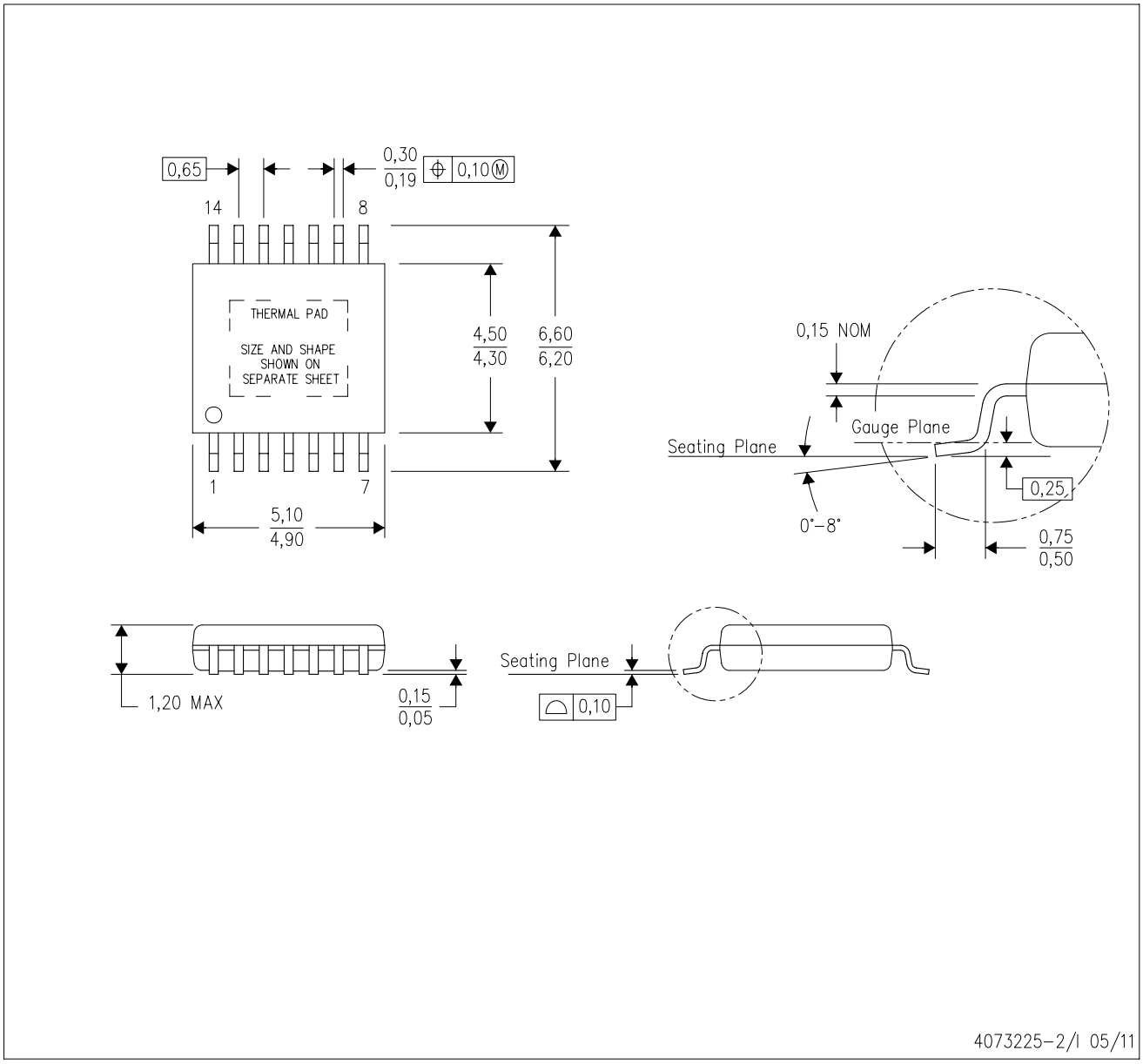


Center Power Pad Solder Stencil Opening		
Stencil Thickness	X	Y
0.1mm	2.5	2.65
0.127mm	2.31	2.46
0.152mm	2.15	2.3
0.178mm	2.05	2.15

4207609–2/W 09/15

NOTES:
A. All linear dimensions are in millimeters.
B. This drawing is subject to change without notice.
C. Customers should place a note on the circuit board fabrication drawing not to alter the center solder mask defined pad.
D. This package is designed to be soldered to a thermal pad on the board. Refer to Technical Brief, PowerPad Thermally Enhanced Package, Texas Instruments Literature No. SLMA002, SLMA004, and also the Product Data Sheets for specific thermal information, via requirements, and recommended board layout. These documents are available at www.ti.com <http://www.ti.com>. Publication IPC–7351 is recommended for alternate designs.
E. Laser cutting apertures with trapezoidal walls and also rounding corners will offer better paste release. Customers should contact their board assembly site for stencil design recommendations. Example stencil design based on a 50% volumetric metal load solder paste. Refer to IPC–7525 for other stencil recommendations.
F. Customers should contact their board fabrication site for solder mask tolerances between and around signal pads.

PWP (R–PDSO–G14) PowerPAD™ PLASTIC SMALL OUTLINE



NOTES:
A. All linear dimensions are in millimeters.
B. This drawing is subject to change without notice.
C. Body dimensions do not include mold flash or protrusions. Mold flash and protrusion shall not exceed 0.15 per side.
D. This package is designed to be soldered to a thermal pad on the board. Refer to Technical Brief, PowerPad Thermally Enhanced Package, Texas Instruments Literature No. SLMA002 for information regarding recommended board layout. This document is available at www.ti.com <http://www.ti.com>.
E. See the additional figure in the Product Data Sheet for details regarding the exposed thermal pad features and dimensions.
E. Falls within JEDEC MO–153

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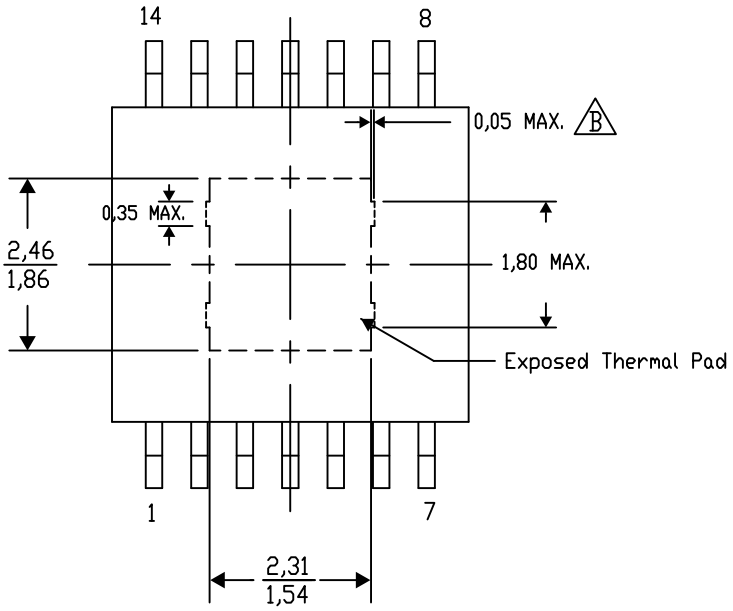
PWP (R–PDSO–G14) PowerPAD™ SMALL PLASTIC OUTLINE

THERMAL INFORMATION

This PowerPAD™ package incorporates an exposed thermal pad that is designed to be attached to a printed circuit board (PCB). The thermal pad must be soldered directly to the PCB. After soldering, the PCB can be used as a heatsink. In addition, through the use of thermal vias, the thermal pad can be attached directly to the appropriate copper plane shown in the electrical schematic for the device, or alternatively, can be attached to a special heatsink structure designed into the PCB. This design optimizes the heat transfer from the integrated circuit (IC).

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The exposed thermal pad dimensions for this package are shown in the following illustration.



Top View

Exposed Thermal Pad Dimensions

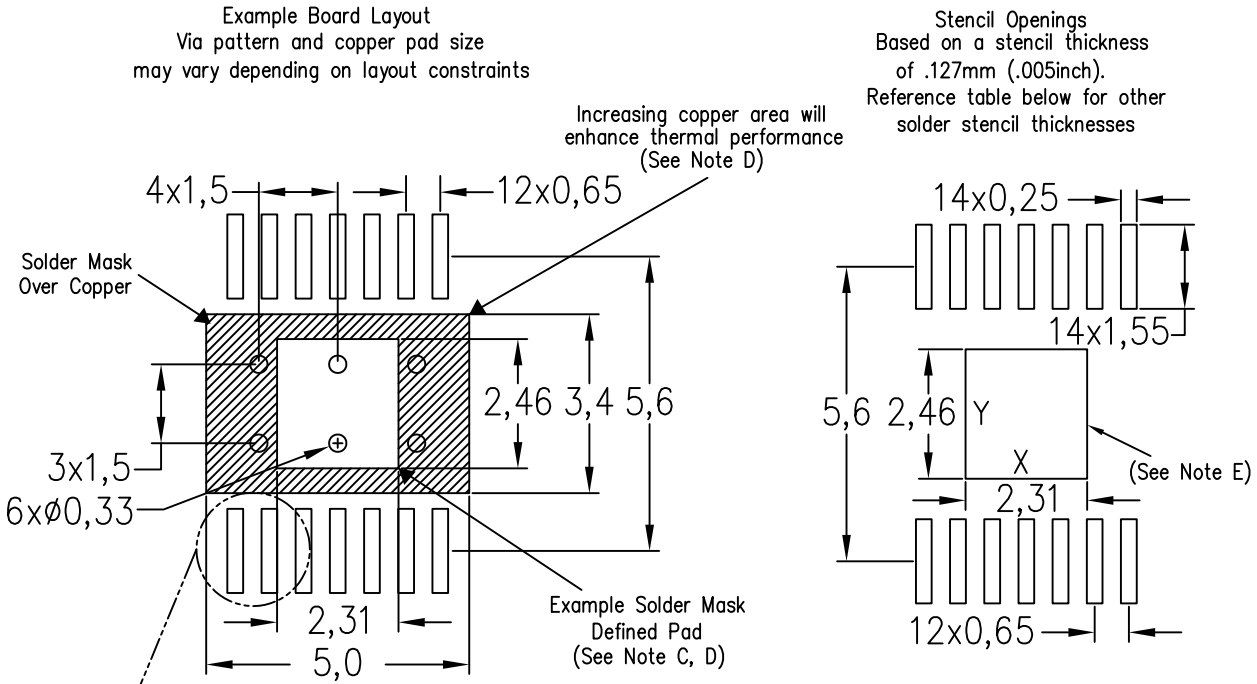
4206332–44/AO 01/16

NOTE: A. All linear dimensions are in millimeters

B. Exposed tie strap features may not be present.

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PWP (R–PDSO–G14) PowerPAD™ PLASTIC SMALL OUTLINE



Center Power Pad Solder Stencil Opening		
Stencil Thickness	X	Y
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0.127mm	2.31	2.46
0.152mm	2.15	2.3
0.178mm	2.05	2.15

4207609–2/W 09/15

NOTES:
A. All linear dimensions are in millimeters.
B. This drawing is subject to change without notice.
C. Customers should place a note on the circuit board fabrication drawing not to alter the center solder mask defined pad.
D. This package is designed to be soldered to a thermal pad on the board. Refer to Technical Brief, PowerPad Thermally Enhanced Package, Texas Instruments Literature No. SLMA002, SLMA004, and also the Product Data Sheets for specific thermal information, via requirements, and recommended board layout. These documents are available at www.ti.com <http://www.ti.com>. Publication IPC–7351 is recommended for alternate designs.
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F. Customers should contact their board fabrication site for solder mask tolerances between and around signal pads.

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